

Toward Car Parking Wellbeing Index.

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ABSTRACT

With the actual development in urban life by the emerging trends around the smart city, along with cars yet considered as the primary source for commuting by city visitors, the parking infrastructures and parking lots management recreates a role in providing an easy mobility service with the sustainability of land use for the city. This ultimately impacted the driver well being. This work provides a foundation for measuring the Parking Wellbeing Index as a tailored refocus of the sustainability society index that takes into account the car parking experience. The framework makes use of the Occupancy rate of the parking site and traffic flow in the vicinity of the parking site, as well as the environmental aspect, which includes both the business/economical dimension and the Green (nature) dimension. We demonstrated how the developed index could be applied in Oulu City center, where a set of experiments have been conducted and tested. Especially, software architecture has been devised to automatically retrieve traffic and parking occupancy using the City of Oulu open data platform. While land use classification has been used to assess the Green and Business aspects of the parking area. The results could help the municipality better manage the buildings and the landscape in a way to maximize community wellbeing.

CCS CONCEPTS

• General and reference; • Cross-computing tools and techniques; • Empirical studies;

KEYWORDS

car parking, wellbeing, land use, data analytic

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1 INTRODUCTION

The interconnection between transportation and wellbeing is of increasing importance to both researchers and policy-makers due to the impact of wellbeing on road safety and driver behavior [11].

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Although research about the direct and indirect links between transport and wellbeing has been relatively scarce to create a comprehensive evidence base, there are strong arguments that testify of this relationship where transport infrastructure is found to influence wellbeing levels by the extent to which it facilitates social connectedness and promotes social inclusion [6]. Transport noise and traffic are also found to cause stress and various health conditions (wellbeing). Therefore, the report [6] stressed that by designing roads to facilitate slow traffic, walking, and pleasing aesthetics, transport policies can help to facilitate community support networks, promote physical activity, and create a pleasant living environment that is conducive to good mental health and wellbeing. Boniface et al. [5] reviewed the impact of transport on social interactions and health. Their findings confirm from using a literature-review-based approach that the strong interlink between community severance and quality of life and wellbeing, where the need for novel methods to quantify wellbeing scores at the community level is stressed. Likewise, the increase in urbanization and wealth as part of socio-economic development has led to a substantial increase in the use of private cars as the main transportation mode[4], which, in turn, yields negative environmental consequences due to higher emissions [9] as well as increased traffic jams[13] and reduced parking availability. Although the use of private cars can positively impact wellbeing, the presence of traffic jams, emissions, and parking unavailability can negatively impact individual or community wellbeing. Therefore, a cautious attitude is required when leveraging transportation modalities in a way to ensure citizen and community wellbeing. With the emergence of new technologies and smart city concepts, new opportunities arise to enhance driver satisfaction and users behavior[2] by enabling smart parking lot assignment and better-assisting users in their travel journey [10][12]. For instance, this enables intelligent systems to gather data on parking and offer real-time information about parking occupancy to drivers, as well as generate itineraries that minimize the impact of traffic jams. The design of research protocol examining travel-wellbeing relationships requires a careful understanding of the psychological attributes being studied, where detailed questionnaires are often required to elicit users' preferences and map to the psychological framework. For instance, the Satisfaction with Travel Scale (STS) [7], which is designed explicitly for quantifying travel wellbeing, emotions and moods are represented on a two-dimensional surface: one dimension (pleasure or valence) ranges from pleasure to displeasure or positive to negative, and the other dimension (arousal or activation) ranges from activation to deactivation. Nevertheless, with the UK transportation review [6], the need for the development of automated wellbeing transport evaluation is emphasized. This paper aims to contribute to this goal by utilizing both the smart city data as well as relevant observations about the users' destinations. We confine our study to data collected from Oulu city, where some

studies have already been performed recently. For instance, Arhab et al. [1] analyzed the parking behavior in Finland by scrapping news articles in Finland by utilizing sentiment scores, topics, and organizations in the vicinity of parking lots. Jutila et al. [8] explored the detailed local plans in the Oulu region, focusing on the usage of resilience in three different study areas in Oulu.

Although, in Oulu, the driver has multiple parking choices: on-street, underground, or parking buildings, this study focuses primarily on parking buildings in Oulu city center. This is because these buildings are equipped with technology-based sensors that enable us to collect real-time parking information, which quantifies the parking occupancy at different hours & days. The results are associated with land use purposes around the parking building by exploring the detailed local plans, while traffic information around parking premises is inferred from the traffic on the nearest roads around the parking building.

2 MATERIALS & METHODS

2.1 Parking wellbeing Index

Capitalizing on commonsense reasoning when a car driver is seeking parking lot in city centre, we hypothesize the following regarding wellbeing evaluation $WL(.)$:

H_1 : Wellbeing WL is dependent on occupancy Occ of parking site, Traffic $Traf$ and Environment Env in the vicinity of the parking area.

H_2 : If traffic $Traf$ is high (resp. low), then wellbeing WL is low (resp. high)

H_1 can be rooted back to sustainable society index (SSI) ¹. In this framework, three dimensions are distinguished: Human wellbeing, Environment wellbeing, and Economic wellbeing. In this context, environmental wellbeing refers to three components: Nature & environment, Resources, and Climate Energy. In the context of our study and given the description assigned to parking building by city planners, the environment factor Env around the parking area includes both the green element (corresponding to Nature & environment in SSI classification) and business activity (corresponding to Economic wellbeing in SSI classification).

Although the relationship between traffic and wellbeing is straightforward in H_2 , the impact of occupancy and environment parameters can be debatable. For instance, a systematically low occupancy rate or very high occupancy rate would indicate that the site is either not that interesting to the user or causes frustration because of a lack of parking availability. In both cases, this negatively impacts the wellbeing WL . The preceding provides a framework for a tailored wellbeing Index, referred to Parking Wellbeing Index (PWI), that accommodates our study, alternative to the SSI framework. This is illustrated in Fig. 1.

Following H_1 and H_2 , PWI can be derived for each area by evaluating each factor (parking occupancy, traffic and environment factor). The latter can be, after some normalization step, evaluated on a scale from 1 to 5 (Very bad to very good), then averaging the three resulted factors yields a global factor (normalized in the unit interval) that represents the parking wellbeing index PWI . Due to

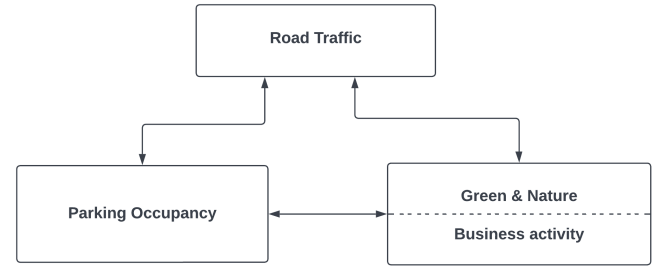


Figure 1: Parking Well Being Index

the lack of relevant information from Oulu Open data service, an approximated method is devised to quantify the three factors of each area based on the best records obtained within the four areas of study. Especially, the environment factor is inferred from the available parking building classification, which reports the presence of economical activities and extent of green area in the vicinity of parking site. In this context, we shall notice the following.

- (1) Traffic: Low traffic represents high level of wellbeing. High traffic reflects poor wellbeing.
- (2) Occupancy rates: very low and very high occupancy reflects poor wellbeing. Good wellbeing can be confirmed by relatively average occupancy with less peak hours.
- (3) Environment: High number green blocks along with the presence of economical activities reflects high wellbeing. The opposite point out average to poor wellbeing.

The preceding can be used as generic guideline to turn heterogeneous raw data of parking occupancy, traffic, business and green presence into normalized 0 to 5 scale evaluation score, which are then used for PWI estimation, see Algorithm 1.

Algorithm 1: Parking wellbeing index estimation

Result: Parking wellbeing index= PWI

Input: Areas= The four area of the study;

Input: Occ= list of occupancy rates in each area;

Input: Env= list of number of VP, A, K in each area;

Input: Traffic= Traffic flow information in each area;

for area in Areas **do**

OccScore= Scale(Occ); // value of occupancy score

EnvScore= Scale(Env); // value of environment score

TrafficScore= Scale(Traffic); // value of traffic score

$PWIndex = \frac{OccScore+EnvScore+TrafficScore}{MaxPoints=15}$

end

2.2 Data collection and software architecture

2.2.1 Parking building classification. Figure 3 shows the surrounding land use categories with their corresponding colors and notation based on detailed local plans with an appropriate distance of 300m

¹https://ec.europa.eu/environment/beyond_gdp/download/factsheets/bgdp-ve-ssi.pdf

around the parking site, in accordance with Finnish Land Use and Building Act. Furthermore, the description of the parking building in terms of green, residential, and business content of the vicinity is illustrated in Table ??.

Basically, each parking building has its own notation², color³, and significance by following marking and regulations of the ministry of the Environment.

2.2.2 Automatic parking data collection. Both parking buildings and traffic data used in this work are collected from Oulu open data platform. The various parking buildings and traffic control stations send data such as free parking space availability and traffic flow information to city servers, which can then be accessed through a GraphQL interface. Due to the nonavailability of historical data, we decided to run a code daily to collect parking buildings data and store it in our local database. Depending on the parking building, parking status and server status, which are updated between 5 to 20 minutes only, so we automate our code to send a request to fetch the data from the servers each half an hour, from 10 AM to 08 PM. The data was mainly collected between 27 October to 10 December 2021. The collected data contains different attributes relevant to our study.

Figure 2 presents a high-level diagram of data collection and overall methodology.

Parking data include the maximum capacity of the parking site, real-time status, timestamp, latitude & longitude of the building, and the number of available parking lots. To calculate the Occupancy score at each timestamp, we subtract the available parking lots from the maximum capacity. A detailed statistical analysis follows this operation to quantify the parking occupancy according to the hourly and day rate. Table 2 presents the parking buildings' names, locations, previous capacity, and newly updated capacity.

To evaluate the car traffic flow in the vicinity of the parking area, we first identified the location of the roads within 300 meters from the parking building location and retrieved the traffic flow at each of these roads. This requires retrieving attributes such as geometry for road coordinates in a multi-string format and their traffic flow information together with the associated timestamp. Especially, the geometry attributes are used to extract the latitudes and longitudes and then identify the nearest routes for each parking based on the Haversine formula that helps us to determine the distance between points on the surface given their longitudes and latitudes [3]. The formula is expressed by Bellow with r equal to 6371 and represents the earth's radius; $dlat$ and $dlon$ represent the difference between two latitudes and longitudes, respectively.

$$Hav = \left(\sin \left(\frac{dLat}{2} \right) \right)^2 + \cos(lat1) \cdot \cos(lat2) \cdot \left(\sin \left(\frac{dLon}{2} \right) \right)^2$$

²The notation follows an official land-use marking scheme where the first letter of each notation provides the main and the wide usage of land use significance (which is more crucial in our study), while the second/third letter of each marking shows the exact usage of the corresponding building. For instance, A: Residential building, and AK: Apartment Building

³Similarly to the notation, each land use category has its own colors following the official land-use marking. For instance, A = Block area of residential buildings = color is brown, light. K = Block area of business and office buildings = color orange-red.

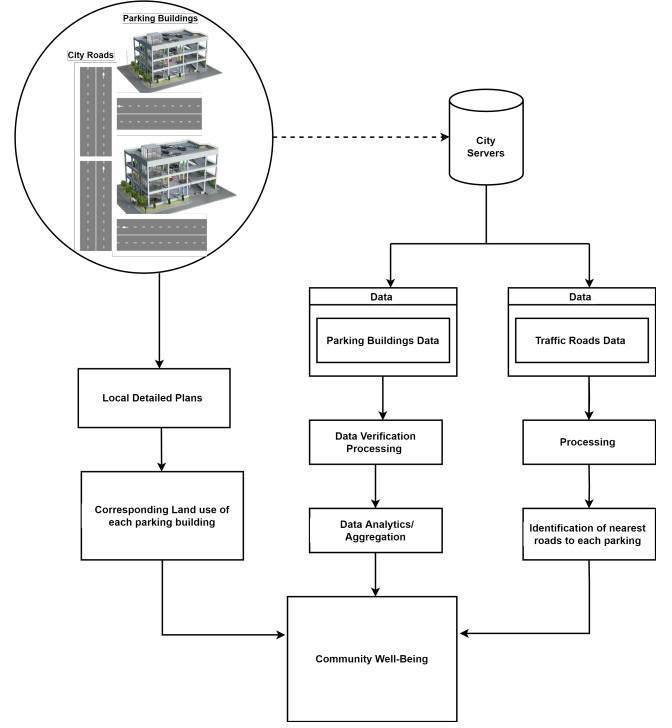


Figure 2: High level diagram of the methodology.

$$Distance = 2 \cdot r \cdot \arcsin \left(\sqrt{Hav} \right)$$

Overall, the methodology for assessing the parking wellbeing index involves i) retrieving the parking occupancy, ii) calculating the traffic flow by averaging road traffic in the vicinity of the parking area at a radius of 300 meters, iii) utilizing the landscape classification provided in Table ?. In the latter, we mainly restricted to the existence of parks (Green wellbeing component) or businesses (economic wellbeing component).

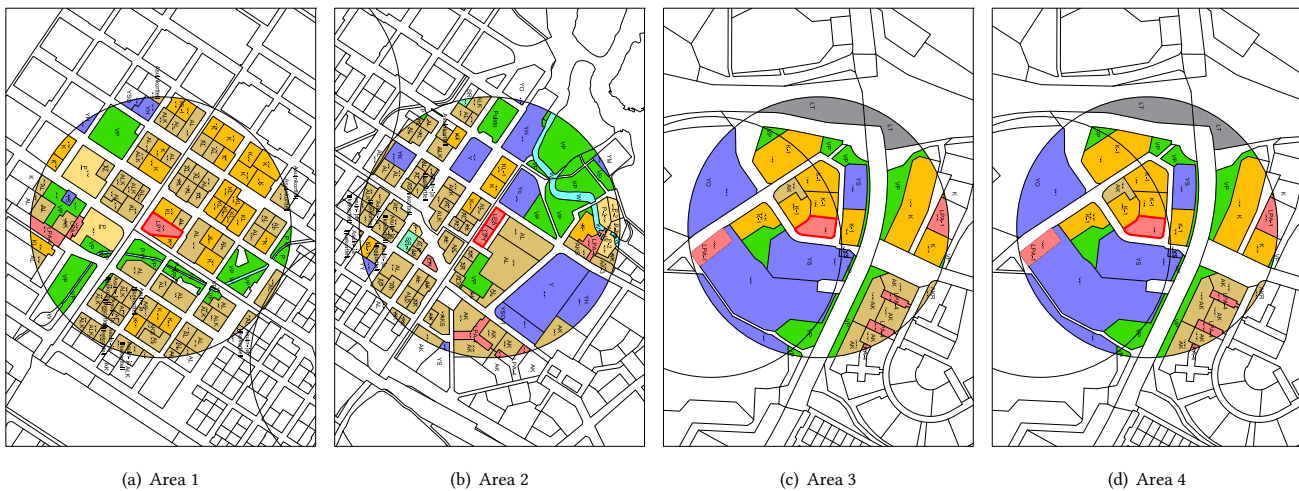
3 RESULTS AND DISCUSSION

First, the data from the four different parking buildings is analyzed. Fig. 4 presents the resulting data from both AREA 1 and 2 parking buildings. And Fig. 5 shows the data from AREA 3 and AREA 4. In addition, Fig. 3 display the land use purpose of each area.

In area 1, the dominant land categories are residential zones with some blocks that include both commercial and office work. The other substantial class is more business oriented, such as commercial blocks and offices, while eight blocks represent parks. Looking at the corresponding parking data of the area, we can see Saturdays have the highest rates of parked cars, up to 97% of total maximum capacity. For other days, we can still notice high occupancy on all remaining days, with the lowest on Fridays at around 89%. Individuals, who are considered residents, frequently use the site to park their cars, which can justify the high number of parked vehicles every day of the week. Besides, commercial and business sites attract more people on Saturdays, which could trigger the peak occupancy rate on that specific day. We also notice that busy hours in parking Area 1 are between 13h to 15h with a maximum of around 14h and

Table 1: Land use categories for the different parking buildings.

Category	land use	Building
AREA 1		
A in 52 blocks	A /AL	Residential / Residential, commercial, and office.
K in 28 blocks	K / KL	- Business and office / Commercial .
V in 8 blocks	VP	- Park.
Other	P / W	- Service building/ Catchment area.
AREA 2		
A in 38 blocks	AL/AK/ ALK/ AKS/ AP/ Ak-3	- Residential/ Apartment/Small residential houses.
V in 9 blocks	VP	- Park
Y in 6 blocks	Y/YH/ YO/ YSO-1/ YK	- Public/ Administrative and office/ teaching activities/ Social services and health care/Churches and parishes.
K in 6 blocks	K/KL/ KL-1	- Business and office/ Commercial.
Other	P/P-2	- Service buildings.
AREA 3		
Y	YS/ YO	- Serving social services and health care/ serving teaching activities.
K	K, K-1	- Business and office buildings.
A	AK	- Apartment buildings.
Other	LPA-4 near AK/ VP/ LT	- Parking spaces/ Park/ Public road area.
AREA 4		
Y	YS/ YO/YO-1/ YK	- Social services and health care/ teaching activities./Churches and other parishes.
K	K-1	- Commercial buildings.
V	VP	- Park.
Other	LPA/ AK/P-4	- Parking places/ Apartment buildings./Service buildings.

**Figure 3: Factual overview of land use purpose from the local detailed plans for the areas around the four parking building.**

very low after 17h. These outcomes could be due to the businesses in the zone where people utilize the parking facility more often when businesses offices are open. Moreover, the traffic flow is good on different days and hours.

In Area 2, we notice that most blocks are relatively residential buildings with 9 blocks of parks and a modest presence of public and administrative blocks. For the corresponding parking data in this zone, we can see that Weekends (Saturday and Sunday) along with

Fridays get a considerable amount of parking to 86% of total capacity on Saturday. For the lower parking days, we can see that Mondays have the lowest parking occupancy rates, with around 65%. These rates are not a surprise for area 2, considering its substantial orientation to a residential side where people park their cars more on weekends with less commuting to work. Low parking rate on Mondays is due to the fact that people are often at work premises. Moreover, the relatively high parking occupancy on weekdays can

Table 2: Parking buildings with real-time parking status information

Name of Parking Building	Server Max Capacity	Updated Max Capacity
Autosaari (AREA 1)	200	421
Autoheikki (AREA 2)	95	225
Rehapiarkki (AREA 3)	357	357
Sairaalaparkki (AREA 4)	102	272

be justified by parking needs for the usage of the administrations and health facilities in the area. When looking at peak hours in this parking site, we can see that the extremes occur at night after 18h, and this confirms and matches the previous conclusion about the residential character of the site, which is characterized by a good and a normal traffic flow near the parking building and a good amount of parks nearby.

For Area 3, the land use categories are more centered around buildings with teaching and health orientation by considering the extensive surface of the land. It also has 8 blocks functioning as parks and a small presence of apartment buildings and offices/businesses. So, we can view this area as a more social service for the community. When looking at the corresponding parking data, we notice that the parking is more useful during all weekdays from Friday 82% to a maximum of 92% or 95% on the remaining weekdays. A lower number of parked cars are located on the weekends in the range 46% to 40%. For the pinnacle hours, we can notice that parking peaks occur between 10h to 14h, and meager counts start after 17h. These results coincide with the land use orientation in this area, with more individuals using the parking facility during the hours or days when social services open for service. So we can conclude that most of the parked cars in the parking site were visitors in that area, and only fewer residents use the parking building, which is also indicated by the light parking count, night hours, and weekends. Finally, the traffic data near the parking buildings was good during the different days and hours.

The fourth and last area in our study, the social class, has the most land use with health, teaching, and parishes services with a large green landscape. Four blocks were reserved for parks, and other minor land use types are commercial blocks and apartments. It reveals that Area 3 and Area 4 have similarities in their parking data. For instance, weekdays are characterized by high parking occurrence in Area 4 from 85% to 89%, with low parking counts on weekends between 77% to 78%. The pinnacle hours in this parking mainly occur between 11h to 14h with a peak of 13h, and meager parking counts start from 16h and beyond.

After the analysis, Table 3 provides the the best area for community wellbeing in descending order.

The results observed from the 4 areas and their corresponding parking data clearly show the effect of the land use purpose on the parking occupancy rates. We detected a rise in parking counts with a higher number of peaks in occupancy in Area 1 and Area 2. The pinnacle hours differ from one place to another depending on the dominant land use from business, residential and social aspects. The residential and business areas need more daytime parking, as highlighted in the above results. Still, one can observe

that Areas 3 and 4 have few (LPA) blocks which refer to parking places close to parking buildings that were missing in zones 1 and 2. So adding a few LPA blocks in Area 1 would be interesting for some users and may reduce the very high occupancy rate in that area. We should also mention the impact of the Covid_19 on the number of parked cars, especially in Areas 1 and 2 (more resident oriented than Areas 3,4), along with the city's recommendations for remote work; this certainly had an effect on the number of parked cars in the building. Lastly, during data collection, Oulu is a city characterized by challenging weather conditions. People often park their cars in underground or parking buildings, which may increase parking demands for such facilities in the wintertime. Lastly, Table 3 presents the classification of the four areas in descending order; all the areas can be considered suitable in the wellbeing index because of the availability of residential, business, and leisure areas around the parking. But we can classify area 2 as the most attractive one among the four with good traffic, the highest number of Parks, and fewer daily rush occupancy hours. On the other hand, despite its social orientation, area 4 is considered the least attractive among the four areas with fewer parks.

4 CONCLUSION

In this work, an association between community well being and car parking modalities was made, and a framework for the car parking wellbeing index has been proposed, which provides a more tailored evaluation of the general sustainability society index. The framework makes use of the Occupancy rate of the parking site and traffic flow in the vicinity of the parking site, as well as the environmental aspect, which includes both the business/economical dimension and the Green (nature) dimension. We demonstrated how the framework could be applied in Oulu City center, where an experiment has been conducted and tested in four different parking sites. For this purpose, software architecture has been developed to automatically gather traffic and occupancy rates from the open data platform provided by city services, while land use classification of the building was used to assess the Environment attribute of wellbeing. The results showed the vital importance of the land use surrounding area in shaping parking occupancy. Especially, four parking buildings and areas were considered for different land use purposes, from social to residential and business/commercial. Moreover, our results highlighted the demand for small extra or multi-function parking in some areas with more residential. In addition, covid_19 possible impact was stressed during the discussion as a factor that has impacted the occupancy of the parking buildings due to city recommendations and motivation for remote work.

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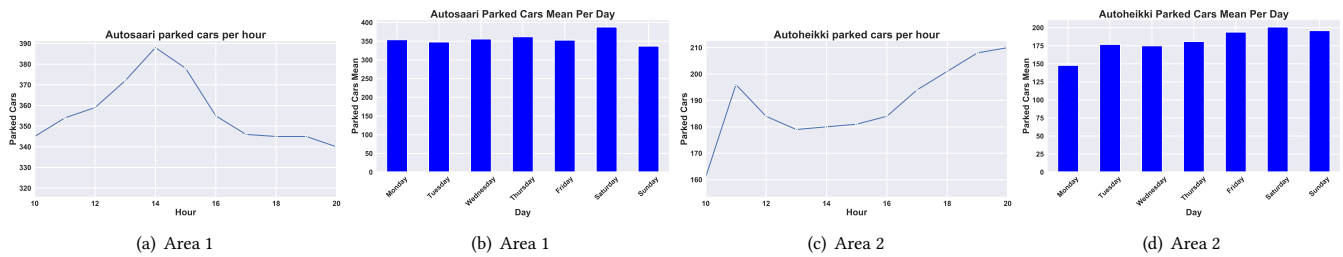


Figure 4: (a) AREA 1 mean parked cars per hour and (b) AREA 1 mean of parked cars per day. (c) AREA 2 mean parked cars per hour and (d) AREA 2 mean of parked cars per day.

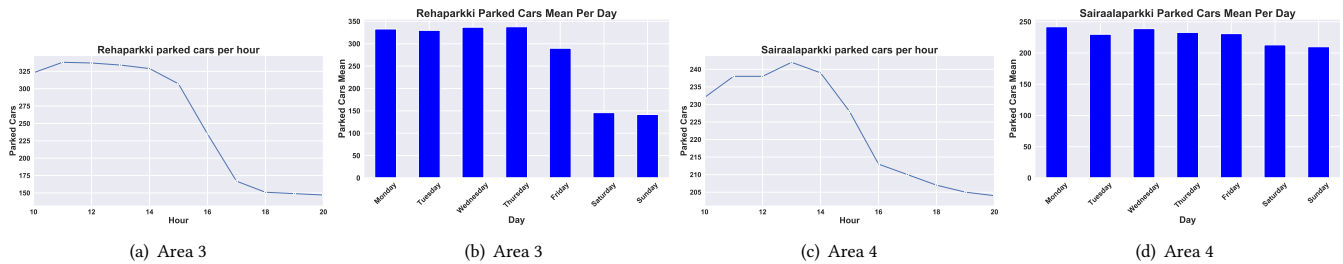


Figure 5: (a) AREA 3 mean parked cars per hour and (b) AREA 3 mean of parked cars per day. (c) AREA 4 mean parked cars per hour and (d) AREA 4 mean of parked cars per day.

Table 3: Classification in descending order of the different areas based of highest wellbeing

AREA	Traffic	Blocks of VP	Dominant land use	highest occupancy per day	highest occupancy hours
Area 2	Good	9	Residential	86%	Night hours
Area 1	Good	8	Residential / Business	97%	1 hours
Area 3	Good	8	Social	92% to 95%	4 hours
Area 4	Good	4	Social	85% to 89%	4 hours

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